THE SHEAFY TRACE MAP IS AN ISOMORPHISM ON RSHEAFHOM

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Sheafy trace equation (equation-sheafy-trace, canonical tag 0GLB). cited at statement line 4151. The map whose being an isomorphism is claimed.
- Adjointness of pullback and pushforward on sites (Cohomology on Sites, Lemma sites-cohomology-lemma-adjoint, canonical tag 07A6). cited at proof lines 4180-4181. First equality: Hom_Y(M, Rf_*(-)) = Hom_X(f^{-1}M, -).
- Tensor internal-Hom adjunction on sites (Cohomology on Sites, Lemma sites-cohomology-lemma-internal-hom, canonical tag 08J9). cited at proof lines 4182-4183 and 4187-4188. Second and fifth equalities: tensor with internal-Hom adjunction on X and on Y.
- Projection formula for Rf_! (Lemma lemma-projection-formula-Rf-lower-shriek, canonical tag 0GL5). cited at proof lines 4185-4186. Fourth equality, the important step: Rf_!(f^{-1}M tensor L) = M tensor Rf_!L.

Source reference. Original equation 0GLB: sources/stacks-chow/tags.txt, source lines 20716–20716. Cited at statement line 4151: the map whose being an isomorphism is claimed.

Source reference. Original lemma 07A6: sources/stacks-chow/tags.txt, source lines 9296–9296. Cited at proof lines 4180-4181: first equality, Hom_Y(M, Rf_*(-)) = Hom_X(f^{-1}M, -).

Source reference. Original lemma 08J9: sources/stacks-chow/tags.txt, source lines 10842–10842. Cited at proof lines 4182-4183 (second equality on X) and proof lines 4187-4188 (fifth equality on Y).

Source reference. Original lemma 0GL5: sources/stacks-chow/tags.txt, source lines 20710–20710. Cited at proof lines 4185-4186: the fourth equality, which the author flags as the important step.

Date: stacks-project:more-etale.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.2. *Let $f : X \to Y$ be a separated finite type morphism of quasi-compact and quasi-separated schemes. Let Λ be a torsion ring. For every $K \in D(Y_{étale}, \Lambda)$ and $L \in D(X_{étale}, \Lambda)$ the map (0GLB)*

$$Rf_* R\mathcal{H}om_\Lambda(L, Rf^!K) \longrightarrow R\mathcal{H}om_\Lambda(Rf_!L, K)$$

is an isomorphism.

Proof. To prove the lemma we have to show that for any $M \in D(Y_{étale}, \Lambda)$ the map (0GLB) induces an bijection

$$\mathrm{Hom}_Y(M, Rf_* R\mathcal{H}om_\Lambda(L, Rf^!K)) \longrightarrow \mathrm{Hom}_Y(M, R\mathcal{H}om_\Lambda(Rf_!L, K))$$

To see this we use the following string of equalities

$$\begin{aligned}
\mathrm{Hom}_Y(M, Rf_* R\mathcal{H}om_\Lambda(L, Rf^!K)) &= \mathrm{Hom}_X(f^{-1}M, R\mathcal{H}om_\Lambda(L, Rf^!K)) \\
&= \mathrm{Hom}_X(f^{-1}M \otimes^{\mathbf{L}}_\Lambda L, Rf^!K) \\
&= \mathrm{Hom}_Y(Rf_!(f^{-1}M \otimes^{\mathbf{L}}_\Lambda L), K) \\
&= \mathrm{Hom}_Y(M \otimes^{\mathbf{L}}_\Lambda Rf_!L, K) \\
&= \mathrm{Hom}_Y(M, R\mathcal{H}om_\Lambda(Rf_!L, K))
\end{aligned}$$

The first equality holds by Cohomology on Sites, Lemma 07A6. The second equality by Cohomology on Sites, Lemma 08J9. The third equality by construction of $Rf^!$. The fourth equality by Lemma 0GL5 (this is the important step). The fifth by Cohomology on Sites, Lemma 08J9. □